Telecommunications
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TIA STANDARD

Project 25

Link Control Word Formats and Messages Addendum 1- Location on PTT

TIA-102.AABF-D-1
(Addendum to TIA-102.AABF-D)

December 2023

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Foreword

(This foreword is not part of this addendum.)

This document has been submitted to APCO/NASTD/FED by the Telecommunications Industry Association (TIA), as provided for in a Memorandum of Understanding (MOU) dated December 1993. That MOU provides that APCO/NASTD/FED will devise a Common System Standard for digital public safety communications (the Standard), and that TIA shall provide technical assistance in the development of documentation for the Standard.

This document has been developed with inputs from the APCO/TIA Project 25 Interface Committee (APIC), the Trunking and Conventional Control Subcommittee (TR-8.10), and TIA Industry members.

This document describes enhancements to the link control word formats and messages in support of Location on PTT functionality utilizing the Common Air Interface.

As other documents describing the conventional and trunking protocols are developed, this document will be subject to change. All reasonable efforts have been made to ensure accuracy, but it should be understood that conformance test remain to be developed and proven.

The symbol “P25” or the wording “Project 25” appearing on the cover sheet of this document when published or in the title of any TIA-102 document or Telecommunications Service Bulletin (TSB) referenced herein, that indicates the APCO/NASTD/FED Project 25 Steering Committee has adopted the document as part of the Standard. The appearance of the abbreviation “P25” or the wording “Project 25” on the cover sheet of this document or in the title of any document referenced herein does not limit the applicability of the information contained in this document to “P25” or “Project 25” implementations exclusively.

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ADDENDUM INTRODUCTION

The purpose of this addendum is to update information contained in TIA-102.AABF-D, "Project 25 Link Control Word Formats and Messages". Details are provided in the "Addendum Scope" section below.

Addendum Scope

This addendum provides the following enhancements to the Project 25 Link Control Word Formats and Messages (TIA-102.AABF-D) standard:

1. Adds new messages in support of the Location on PTT functionality.

This document only includes information relevant to the scope as described above. The reader is referred to TIA-102.AABF-D for all other pertinent information. The next complete revision of TIA-102.AABF-D will incorporate the changes contained in this addendum.

Note that section numbers identified in this addendum directly translate to section numbers in the TIA-102.AABF-D, Project 25 Link Control Word Formats and Messages.

Addendum Revision History

Version	Date	Description
TIA-102.AABF-D-1	Dec 2023	Approved for publication.

References

The following standards contain provisions which, through reference in this text, constitute provisions of this Standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this Standard are encouraged to investigate the possibility of applying the most recent editions of the standards indicated below. TIA maintains registers of currently valid national standards published by them.

[BAJC] Project 25 Tier 2 Location Services Specification, TIA-102.BAJC-B, TIA, March 2019 {NORMATIVE}

[BBAD] Project 25 Two-Slot TDMA MAC Layer Messages, TIA-102.BBAD-A, TIA, November 2019 {NORMATIVE}

Addendum changes to TIA-102.AABF-D, Link Control Word Formats and Messages, begin here.

[Changes begin as identified below. No changes from the existing text before this clause]

7.3 LC Messages

7.3.35 MFIDA4 GPS Update Part A

This LC is used inbound during an FDMA call in a group or regroup to provide a portion of a subscriber unit's GPS location reading. The location of the subscriber is determined by combining this LC with the GPS Update Part B. Represenation of coordinates in the updates include indications for hemisphere and provide magnitudes in DDMM.mmmm format for latitude and DDDMM.mmmm format for longitude.

<table>
<tr><td>Octet 0</td><td>P</td><td>SF</td><td colspan="6">LCO</td></tr>
<tr><td>1</td><td colspan="8">Manufacturer ID</td></tr>
<tr><td>2
3</td><td colspan="8">Latitude .mmmm</td></tr>
<tr><td>4</td><td>Lat H</td><td colspan="7">Latitude MM</td></tr>
<tr><td>5</td><td colspan="8">Latitude DD</td></tr>
<tr><td>6
7</td><td colspan="8">Longitude .mmmm</td></tr>
<tr><td>8</td><td>Lon H</td><td colspan="7">Longitude MM</td></tr>
<tr><td>Bit</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
</table>

LCO = 42 (%10,1010), or 44(%10,1100)
P = 0, **SF** = 0

LC Opcode: 42(%10,1010) for odd-numbered updates (1st, 3rd, 5th update).
44(%10,1100) for even-numbered updates (2nd, 4th,6th update).

For odd-numbered updates, including the first update, the radio uses LCO=42 for Part A and LCO=43 for Part B.

FOR even-numbered updates, the radio uses LCO=44 for Part A and LCO=45 for Part B.

Manufacturer ID: Set to $A4

Latitude .mmmm: Two octet field conveying the decimal part of minutes latitude in ten thousandths of minutes. Minutes latitude can be reconstructed from the packet as follows: Minutes latitude = Latitude MM + (Latitude .mmmm / 10,000).

Lat H: Hemisphere of latitude
%0 = North
%1 = South

Latitude MM: 7-bit field conveying integer part of minutes latitude in MM.mmm format

Latitude DD: Degrees of latitude, with LSB = 1 degree.

Longitude .mmmm: Two octet field conveying the decimal part of minutes longitude in ten thousandths of minutes. Minutes longitude can be reconstructed from the packet as follows: Minutes longitude = Longitude MM + (Longitude .mmmm / 10,000).

Lon H: Hemisphere of longitude
%0 = East
%1 = West

Longitude MM: 7-bit field conveying integer part of minutes longitude in MM.mmmm format.

7.3.36 MFIDA4 GPS Update Part B

This LC is used inbound during an FDMA call on a group or regroup to provide a portion of a subscriber unit's GPS location reading. The location of the subscriber is determined by combining this LC with the MFIDA4 GPS Update Part A. Represenation of coordinates in the updates include indications for hemisphere and provide magnitudes in DDMM.mmmm format for latitude and DDDMM.mmmm format for longitude.

Octet	7	6	5	4	3	2	1	0
0	P	SF	LCO					
1	Manufacturer ID							
2	Longitude DDD							
3 4	Time - lower							
5	Heading - lower							
6	Velocity - lower							
7	Velocity - upper				Heading - upper			
8	T	E	Quality		Number of Satellites			
Bit	7	6	5	4	3	2	1	0

LCO = 43 (%10,1011),
or 45(%10,1101)
P = 0, **SF** = 0

LC Opcode:	43 (%10,1011) for odd-numbered updates (1^{st}, 3^{rd}, 5^{th} update). 45 (%10,1101) for even-numbered updates (2^{nd}, 4^{th},6^{th} update). For odd-numbered updates, (including the first update) the radio uses LCO=42 for Part A and LCO=43 for Part B. For even-numbered updates, the radio uses LCO=44 for Part A and LCO=45 for Part B.
Manufacturer ID:	Set to $A4
Longitude DDD:	Longitude in degrees (LSB = 1 degree)
Time - lower:	Lower 16 bits of a 17-bit representation of number of seconds since midnight. $1FFFF indicates time is unknown. (See T in this message) NOTE: The receiver is expected to fill in the date based upon the day the location update is received and the time included in the location update.
Heading - lower:	Heading of the subscriber unit, in tenths of degrees. This is the lower 8 bits of heading. See Heading – upper.
Velocity - lower:	Velocity of the subscriber unit, reported in tenths of knots. This is the lower 8 bits of velocity. See Velocity – upper.
Velocity – upper:	Velocity of the subscriber unit, reported in tenths of knots. This is the upper 4 bits of velocity. See Velocity – lower.
Heading – upper:	Heading of the subscriber unit, in tenths of degrees. 0 indicates North. This is the upper 4 bits of heading. See Heading – lower.
T	17^{th} (MSB) bit of Time (see Time - lower in this message)
E	Indicates SU emergency status. %0 SU not in emergency %1 SU in emergency.
Quality	GPS Quality indicator %00 – No GPS fix %01 – GPS fix established %10 – Differential GPS fix established

%11 - GPS not enabled or no GPS-capable hardware present.

Number of Satellites: Number of GPS satellites in view

7.3.37 MFID90 Location on PTT Header

This LC is used inbound during a standard group call or MFID90 supergroup call to identify the transmitting unit and its last GPS location reading. The location of a subscriber is provided by combining this LC with the MFID90 Location on PTT message payload LC.

Octet	Bit 7	6	5	4	3	2	1	0
0	P	SF	LCO					
1	Manufacturer ID							
2	TYPE		DV	0	0	0	E	SOP
3	Source Address							
4								
5								
6	Latitude							
7								
8								

LCO = 41 (%00,101001)
P = 0, **SF** = 0

Manufacturer ID: Set to $90

TYPE: Indicates GPS reading type; Set to %00 Point-2D specifies a point (Latitude and Longitude) without altitude information.

DV: Direction valid –
Set to 0 for direction field in LC_MOT_PTT_LOC_PAYLD is not valid.
Set to 1 for direction field in LC_MOT_PTT_LOC_PAYLD is valid.

E bit: Indicates the current emergency transmission state of the call. Set to %1 for emergency transmissions. Set to %0 for non-emergency transmissions.

SOP bit: Equivalent to service options P-Bit. Indicates the current transmission encryption state. Set to %0 for clear calls and set to %1 for encrypted calls.

Source Address: The WUID of the subscriber currently transmitting.

Latitude: The three byte latitude value is determined by first discarding the least significant byte of the four byte LRRP latitude value and then rounding up (incremented) the result if the discarded byte was >= 0x80. Note that the value of the sign bit must be copied from the most significant bit of the LRRP value, to the most significant bit of the three byte value.

The encoding uses 1 bit (the most significant bit of the first octet) to indicate the sign, and 31 bits to encode the unit. To represent an (absolute) latitude value L (where L is specified in degrees), the value is encoded as $((L * 2^{31}) / 90)$. As an exception, the value for (absolute) L=90 is encoded as 2^{31}-1.

Example:
The encoding for latitude (specified in decimal degrees) 12.345345 which is hexadecimal (4 octets) is 118ECD8D. ‘8D’ is truncated in this case.

Refer to **[BAJC]** for LRRP details.

7.3.38 MFID90 Location on PTT Payload

This LC is used inbound during a group call or MFID90 supergroup call to identify the transmitting unit and its last GPS location reading. The location of a subscriber is determined by combining this LC with the MFID90 Location on PTT header LC.

Octet	7	6	5	4	3	2	1	0
0	P	SF	LCO					
1	Manufacturer ID							
2	Longitude							
3								
4								
5	Minute						Second	
6	Direction			Speed				
7	Reserved				CRC-12 over entire payload			
8								

LCO = 42 (%00,101010)
P = 0**, SF** = 0

Manufacturer ID: Set to $90

Longitude: The three byte longitude value is be determined by taking the absolute value of the four byte LRRP longitude value and then discarding the least significant byte. The resulting three byte value will then be rounded up (incremented) if the discarded byte was >= 0x80. Finally the most significant bit of the three byte value will be set if the LRRP longitude value was negative.
The encoding uses a 32-bits 2’s complement binary to represent the longitude. To represent an (absolute) longitude value L (where L is specified in degrees), the value is encoded as $((L * 2^{32}) / 360)$.

Example:
The encoding for longitude (specified in decimal degrees) 24.668866 which is hexadecimal (4 octets) is 118AD47B.

Minute: GPS sample minute resulting from forward adjusted second.
%000000 = minute 0,
%000001 = minute 1,
%000010 = minute 2,
%111011 = minute 59.

Second: GPS sample forward adjusted second.
%00 = 0 seconds,
%01 = 15 seconds,
%10 = 30 seconds,
%11 = 45 seconds.

Direction: The direction in with 45 degrees of resolution.
Set to %000 for N <337.5 to less than 22.5 degrees>,
%001 for NE <22.5 to less than 67.5 degrees >,
%010 E <67.5 to less than 112.5 degrees >,
%011 SE <112.5 to less than 157.5 degrees>,
%100 S <157.5 to less than 202.5 degrees >,
%101 SW <202.5 to less than 247.5 degrees>,
%110 W <247.5 to less than 292.5 degrees>,
%111 NW <292.5 to less than 337.5 degrees >.

Speed: Set to %00000 for 0 meters per second;
%00001 for greater than zero meters per second to less than 2 meters per second;
%00010 for 2 meters per second to less than 4 meters per second;
%11110 for 58 meters/second to less than 60 meters per second;
%11111 for 60 meters/second (134.2 miles/ hour) or greater.

CRC: 12bit CRC. Calculated the same way a 12-bit MAC PDU CRC is calculated per **[BBAD]** section 4.6. CRC computed over the combined MFID90 Location on PTT message LCs beginning with the first octet of the MFID90 Location on PTT Header down to the last bit of the final MFID90 Location on PTT Payload octet up to the CRC field. (Example: A Type %000 MFID90 Location on PTT message computes the CRC-12 over 16.5 octets of payload information.)

7.4 Standard LC Field Definitions

[Added Link Control Opcodes]

Add to Table 2: LCO – Link Control Opcode

LCO Value	Alias Name	Full Name Mnemonic
41 (%00,101001)	LC_MOT_PTT_LOC_HDR	MFID90 Location on PTT message Header
42 (%00,101010)	LC_MOT_PTT_LOC_PAYLD	MFID90 Location on PTT message Payload
42 (%00,101010)	MFIDA4_GPS_A_ODD	GPS Update Part A – Odd numbered update
43 (%00, 101011)	MFIDA4_GPS_B_ODD	GPS Update Part B –Odd-numbered update
44 (%00, 101100)	MFIDA4_GPS_A_EVEN	GPS Update Part A – Even numbered update
45 (%00, 101101)	MFIDA4_GPS_B_EVEN	GPS Update Part B – Even-numbered update

7.5 Link Control Word Usages

[Added Link Control Opcodes]

Add to Table 3: Link Control Word Usage

Conventional Outbound

Conventional Inbound

Trunked Outbound

Trunked Inbound

LC Function	LCO Value	↓	↓	↓	↓
GPS Update Part A – Odd numbered update	42 (%00,101010)	x			
GPS Update Part B – Even numbered update	43 (%00,101011)	x			
GPS Update Part A – Even numbered update	44 (%00,101100)	x			
GPS Update Part B – Odd numbered update	45 (%00,101101)	x			
MFID90 Group Data Header Block (Location on PTT)	41 (%00,101001)	x			
MFID90 Group Data Payload (Location on PTT)	42 (%00,101010)	x			

--- End of Addendum ---

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